

Lab 8 - Working with Tables

Table functions in DAX are powerful tools for analyzing and manipulating data within Power BI or other compatible platforms. They enable users to create new tables, filter existing data, and calculate aggregations efficiently. By leveraging these functions, data professionals can extract meaningful insights and maintain precise control over their datasets, ensuring a more dynamic and customized analytical experience.

Common Table functions

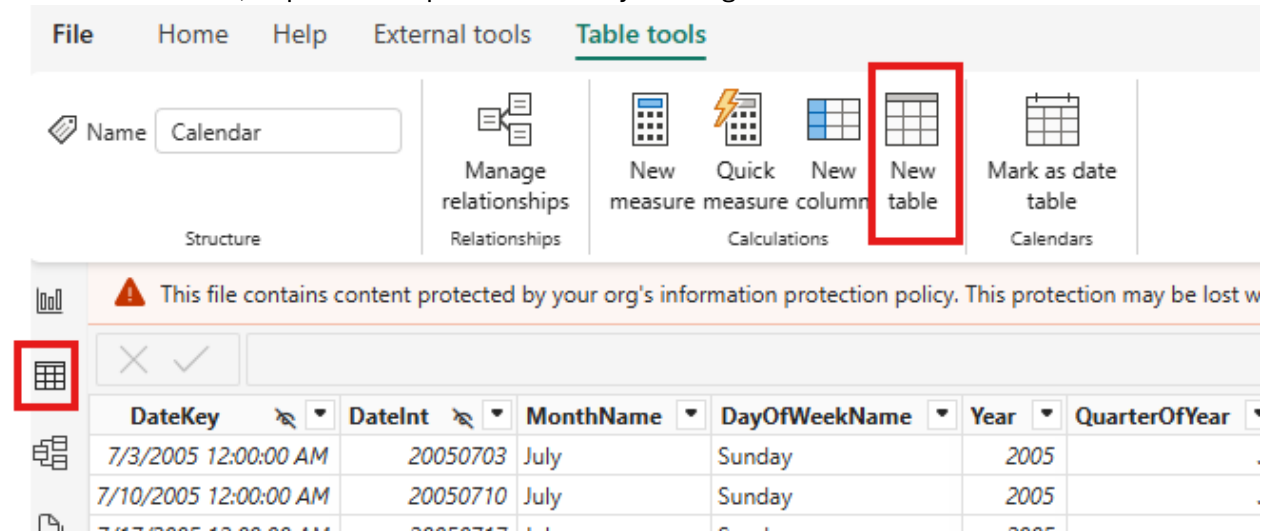
- Filter
- All
- Allexcept
- Values
- Distinct
- Blank row
- Allselected

In this lab we will walk through examples and concepts of working with tables using the above functions.

Section 8.1 - How to duplicate tables

Duplicating a table is as simple as referencing the table name

8.1 – In Table view, duplicate the product table by clicking on “new table” on the table tools menu



8.2 – Enter the following formula in the formula bar to duplicate the product table.

New Product Table = 'Product'

Notice this created a complete duplicate table. All the rows and all the columns.

Section 8.2 - How to modify table rows

FILTER – Modify the rows in a table

What if we want to create a modified version of the existing table by reducing the number of values or rows in the table?

The first table function we will look at is FILTER. [FILTER - DAX Guide](#)

Filter makes a table smaller (filtered on rows)

Filter returns all columns in the table

Filter is the only table expression that filters rows



What if we wanted the Product Table to only list products for the Contoso Brand?

8.3 – Create a new table and enter the following in the formula bar

Product Table with Filter = `FILTER('Product', 'Product'[BrandName]="Contoso")`

Notice this created a filtered version of 'Product' with the [BrandName] column filtered by "Contoso"

You can filter multiple tuples (multiple column filter arguments)

8.4 – Update the formula from step 3 with the following"

Product Table with Filter = `FILTER('Product', 'Product'[BrandName]="Contoso" || 'Product'[BrandName] = "Fabrikam")`

Notice we use a DAX operator (|| - The logical OR) to create a filter on one column for two different values.

8.5 – This same table can be written using another DAX Operator – IN. IN returns TRUE if a value exists or is contained in a table. Update the formula from step 4 with the following:

```
Product Table with Filter = FILTER('Product', 'Product'[BrandName] IN {"Contoso", "Fabrikam"})
```

This creates an unnamed table with 2 values in it – Contoso and Fabrikam – and then uses that unnamed table to filter the table being created.

Instead of selecting items to include in the filter, we can select items to not include or remove from the table.

8.6 – Update the formula in step 5 with the following –

```
Product Table with Filter = FILTER('Product', 'Product'[BrandName] <> "Contoso")
```

Notice it included all the values except for Contoso

You can filter with expressions

8.7 – Update the formula with the following

```
Product Table with Filter = FILTER('Product', 'Product'[UnitCost] >= 100)
```

8.8 – We can apply filters on multiple columns. Update the formula with the following

```
Product Table with Filter = FILTER('Product', 'Product'[UnitCost] >= 100 && 'Product'[BrandName] IN {"Contoso", "Fabrikam"})
```

This created a filter where UnitCost is over \$100 AND BrandName is either Contoso or Fabrikam

8.9 – If we change the && to OR what does this formula create?

```
Product Table with Filter = FILTER('Product', 'Product'[UnitCost] >= 100 || 'Product'[BrandName] IN {"Contoso", "Fabrikam"})
```

A filtered table for all Contoso and Litware products no matter the UnitCost, and all other BrandNames with UnitCost over \$100.

Section 8.3 - Challenge Questions

Question 1 – Create a version of the product table that only shows products that cost \$100 or more and under \$200.

Question 2 – Create a table that shows a list of products for “Litware” that have a UnitCost of \$100 or more and less than \$200.

Question 3 – Try to create a table using FILTER without a filter argument.

Section 8.4 - Challenge Answers:

Question 1 –

Product Table with Filter = `FILTER('Product', 'Product'[UnitCost]>=100 && 'Product'[UnitCost] <200)`

Question 2 -

Product Table with Filter = `FILTER('Product', 'Product'[UnitCost]>=100 && 'Product'[UnitCost] <200 && 'Product'[BrandName] = "Litware")`

Question 3 –

FILTER requires a filter argument. You can enter 1 for the filter and it will return the whole table.

Section 8.5 – How to modify table columns

SELECTCOLUMNS – Modify the columns in a table

What if we want to create a version of a table that includes all the rows but only some of the columns?

8.10 – Create a version of the ‘Product’ table that only has the [ProductName] and [UnitCost] columns. Enter the following formula when creating a “new table”-

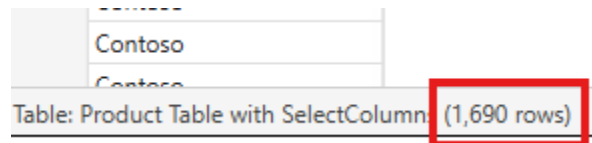
Product Table with SelectColumns =

SELECTCOLUMNS('Product','Product'[ProductName],'Product'[UnitCost])

Notice this new table still returned the 1690 rows from the original product table. It created the columns with no filter on the rows.

8.11 - What happens if we use SELECTCOLUMNS with one column with only a few distinct values in the row? Update the table formula from step 10 with the following -

Product Table with SelectColumns = **SELECTCOLUMNS**('Product','Product'[BrandName]) -- it still returns all the rows even though there is only 9 distinct values.



Contoso
Contoso

Table: Product Table with SelectColumn (1,690 rows)

ALL – Modify tables or columns, deduplicate values and ignore filter context

8.13 - What if we want to return a deduplicated column of distinct values?

We can use the DAX function ALL.

Let's change from creating new tables in the model to testing these functions using DAX Query view. As of the March 2025 release DAX Query view is generally available. If you are on an older version of power bi desktop, it is a preview feature that needs to be enabled in Options and Settings.

From the home ribbon select File-Options and settings-Options then Global-Preview Features and select "DAX Query view".

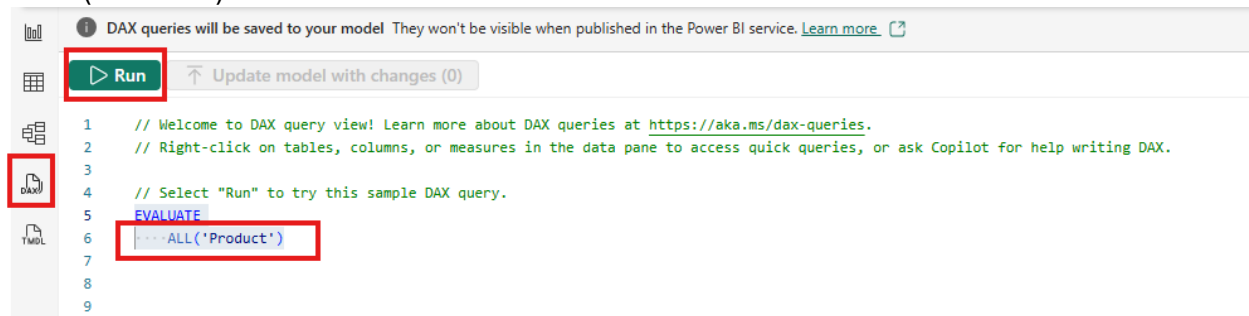
Query view lets you test DAX queries and see the results without creating the table in the model. EVALUATE is a DAX

Generate a query in DAX Query view with the following formula

Create a new table with the following formula the click "Run" –

EVALUATE

ALL('Product')



Notice that ALL with a table name returns the whole table

Results | Result 1 of 1 | Copy

	Product[ProductName]	Product[ProductDescrip...	Product[Manufacturer]	Product[BrandName]	Product[ClassName]	Product[UnitCost]	Product[UnitPrice]	Product[ProductKey]	Product[ProductSubcat...
1	Contoso Wireless Laser ...	Advanced 2.4 GHz cordl...	Contoso, Ltd	Contoso	Economy	10.69	20.96	873	22
2	Contoso Optical Wheel ...	PS/2 mouse, 6 feet mou...	Contoso, Ltd	Contoso	Economy	6.63	13	879	22
3	Contoso Optical Wheel ...	PS/2 mouse, 6 feet mou...	Contoso, Ltd	Contoso	Economy	6.63	13	880	22
4	Contoso Optical Wheel ...	PS/2 mouse, 6 feet mou...	Contoso, Ltd	Contoso	Economy	6.63	13	881	22
5	Contoso Optical Wheel ...	PS/2 mouse, 6 feet mou...	Contoso, Ltd	Contoso	Economy	6.63	13	882	22
6	Contoso Bluetooth Not...	Transceiver-free wireles...	Contoso, Ltd	Contoso	Economy	25.49	50	887	22
7	Contoso Bluetooth Not...	Transceiver-free wireles...	Contoso, Ltd	Contoso	Economy	25.49	50	888	22
8	Contoso Bluetooth Not...	Transceiver-free wireles...	Contoso, Ltd	Contoso	Economy	25.49	50	889	22
9	Contoso Bluetooth Not...	Transceiver-free wireles...	Contoso, Ltd	Contoso	Economy	25.49	50	890	22
10	Contoso ADSL Modem ...	ADSL modem splitter/fil...	Contoso, Ltd	Contoso	Economy	9.13	17.9	833	22
11	Contoso ADSL Modem ...	ADSL modem splitter/fil...	Contoso, Ltd	Contoso	Economy	11.68	22.9	834	22
12	Contoso ADSL Modem ...	ADSL modem splitter/fil...	Contoso, Ltd	Contoso	Economy	26.97	52.9	835	22
13	Contoso USB Wave Mul...	USB wave multimedia k...	Contoso, Ltd	Contoso	Economy	8.11	15.9	836	22

Query 1 | 9 columns, 1,690 rows

8.14 – Generate a new query adding a table and column reference with the following and click “Run”.

EVALUATE

```
ALL('Product'[BrandName])
```

Notice ALL with a column reference returns the unique values from that column.

Results | Result 1 of 1  Copy 

	Product[BrandName]
1	Contoso
2	Wide World Importers
3	Northwind Traders
4	Adventure Works
5	Southridge Video
6	Litware
7	Proseware
8	A. Datum
9	The Phone Company
10	Fabrikam

		 Query 1	
Success (34.8 ms)	Query 1 of 1	Result 1 of 1	1 column, 10 rows

8.15 – If we use ALL with multiple columns it returns a table of unique distinct combos. Generate a query with the following –

EVALUATE

```
ALL('Product'[BrandName],'Product'[Manufacturer])
```

Notice it creates a 2 column table of unique values

Results | Result 1 of 1 Copy

	Product[BrandName]	Product[Manufacturer]
1	Contoso	Contoso, Ltd
2	Wide World Importers	Wide World Importers
3	Northwind Traders	Northwind Traders
4	Adventure Works	Adventure Works
5	Southridge Video	Southridge Video
6	Litware	Litware, Inc.
7	Proseware	Proseware, Inc.
8	A. Datum	A. Datum Corporation
9	The Phone Company	The Phone Company
10	Fabrikam	Fabrikam, Inc.

	Query 1		
ess (49.1 ms)	Query 1 of 1	Result 1 of 1	2 columns, 10 rows

8.16 – Generate a new query adding a third column

EVALUATE

```
ALL('Product'[BrandName], 'Product'[Manufacturer], 'Product'[ClassName])
```

Results | Result 1 of 1 Copy

	Product[BrandName]	Product[Manufacturer]	Product[ClassName]
1	Contoso	Contoso, Ltd	Economy
2	Wide World Importers	Wide World Importers	Economy
3	Northwind Traders	Northwind Traders	Economy
4	Adventure Works	Adventure Works	Economy
5	Southridge Video	Southridge Video	Economy
6	Litware	Litware, Inc.	Economy
7	Proseware	Proseware, Inc.	Economy
8	A. Datum	A. Datum Corporation	Economy
9	The Phone Company	The Phone Company	Economy
10	Fabrikam	Fabrikam, Inc.	Economy
11	Contoso	Contoso, Ltd	Regular
12	Wide World Importers	Wide World Importers	Regular
13	Northwind Traders	Northwind Traders	Regular

Query 1

ess (123.6 ms) Query 1 of 1 Result 1 of 1 3 columns, 29 rows

Notice there a now 3 columns with 29 rows, one row for each unique row combo

ALLEXCEPT – Modify a table by excluding columns

ALL includes columns, **ALLEXCEPT** excludes columns

8.17 – Generate a new query with the below formula

EVALUATE

ALLEXCEPT('Product','Product'[ProductDescription])

Notice this returns all the rows (1690) and only 8 of the columns, excluding 'Product'[ProductDescription]

Results | Result 1 of 1 | Copy

	Product[ProductName]	Product[Manufacturer]	Product[BrandName]	Product[ClassName]	Product[UnitCost]	Product[UnitPrice]	Product[ProductKey]	Product[ProductSubcat...]
1	Contoso Wireless Laser ...	Contoso, Ltd	Contoso	Economy	10.69	20.96	873	22
2	Contoso Optical Wheel ...	Contoso, Ltd	Contoso	Economy	6.63	13	879	22
3	Contoso Optical Wheel ...	Contoso, Ltd	Contoso	Economy	6.63	13	880	22
4	Contoso Optical Wheel ...	Contoso, Ltd	Contoso	Economy	6.63	13	881	22
5	Contoso Optical Wheel ...	Contoso, Ltd	Contoso	Economy	6.63	13	882	22
6	Contoso Bluetooth Not...	Contoso, Ltd	Contoso	Economy	25.49	50	887	22
7	Contoso Bluetooth Not...	Contoso, Ltd	Contoso	Economy	25.49	50	888	22
8	Contoso Bluetooth Not...	Contoso, Ltd	Contoso	Economy	25.49	50	889	22
9	Contoso Bluetooth Not...	Contoso, Ltd	Contoso	Economy	25.49	50	890	22

Query 1 +

ess (326.3 ms) Query 1 of 1 Result 1 of 1 8 columns, 1,690 rows

VALUES – modify a table or column by keeping filter context and deduplicating columns and keeping the blank row

8.18 – Another table modification function is VALUES. Values returns a table or unique values in a column honoring filter context and deduplicating the column values in a column.

Enter the following query

EVALUATE

VALUES('Product')

Notice Values with a table returns the whole table – all columns and all rows

Results | Result 1 of 1 | Copy

	Product[ProductName]	Product[ProductDescrip...]	Product[Manufacturer]	Product[BrandName]	Product[ClassName]	Product[UnitCost]	Product[UnitPrice]	Product[ProductKey]	Product[ProductSubcat...]
1	Contoso Wireless Laser ...	Advanced 2.4 GHz cordl...	Contoso, Ltd	Contoso	Economy	10.69	20.96	873	22
2	Contoso Optical Wheel ...	PS/2 mouse, 6 feet mou...	Contoso, Ltd	Contoso	Economy	6.63	13	879	22
3	Contoso Optical Wheel ...	PS/2 mouse, 6 feet mou...	Contoso, Ltd	Contoso	Economy	6.63	13	880	22
4	Contoso Optical Wheel ...	PS/2 mouse, 6 feet mou...	Contoso, Ltd	Contoso	Economy	6.63	13	881	22
5	Contoso Optical Wheel ...	PS/2 mouse, 6 feet mou...	Contoso, Ltd	Contoso	Economy	6.63	13	882	22
6	Contoso Bluetooth Not...	Transceiver-free wireles...	Contoso, Ltd	Contoso	Economy	25.49	50	887	22
7	Contoso Bluetooth Not...	Transceiver-free wireles...	Contoso, Ltd	Contoso	Economy	25.49	50	888	22
8	Contoso Bluetooth Not...	Transceiver-free wireles...	Contoso, Ltd	Contoso	Economy	25.49	50	889	22
9	Contoso Bluetooth Not...	Transceiver-free wireles...	Contoso, Ltd	Contoso	Economy	25.49	50	890	22

Query 1 +

ess (72.6 ms) Query 1 of 1 Result 1 of 1 9 columns, 1,690 rows

8.19 – Enter the following query for a column with VALUES

EVALUATE

VALUES('Product'[Manufacturer])

Values with a column reference returns the distinct values in referenced column

Results | Result 1 of 1 Copy

	Product[Manufacturer]
1	Contoso, Ltd
2	Wide World Importers
3	Northwind Traders
4	Adventure Works
5	Southridge Video
6	Litware, Inc.
7	Proseware, Inc.
8	A. Datum Corporation
9	The Phone Company
10	Fabrikam, Inc.

Query 1

cess (36.9 ms) | Query 1 of 1 | Result 1 of 1 | 1 column, 10 rows

DISTINCT – modify a table or column by keeping filter context and deduplicating columns and ignoring the blank row

8.20 Enter the following query with DISTINCT

EVALUATE

DISTINCT('Product')

At this point this looks similar to ALL and VALUES – it returns the whole table

Results Result 1 of 1 Copy

	Product[ProductName]	Product[ProductDescrip...	Product[Manufacturer]	Product[BrandName]	Product[ClassName]	Product[UnitCost]	Product[UnitPrice]	Product[ProductKey]	Product[ProductSubcat...
1	Contoso Wireless Laser ...	Advanced 2.4 GHz cordl...	Contoso, Ltd	Contoso	Economy	10.69	20.96	873	22
2	Contoso Optical Wheel ...	PS/2 mouse, 6 feet mou...	Contoso, Ltd	Contoso	Economy	6.63	13	879	22
3	Contoso Optical Wheel ...	PS/2 mouse, 6 feet mou...	Contoso, Ltd	Contoso	Economy	6.63	13	880	22
4	Contoso Optical Wheel ...	PS/2 mouse, 6 feet mou...	Contoso, Ltd	Contoso	Economy	6.63	13	881	22
5	Contoso Optical Wheel ...	PS/2 mouse, 6 feet mou...	Contoso, Ltd	Contoso	Economy	6.63	13	882	22
6	Contoso Bluetooth Not...	Transceiver-free wireles...	Contoso, Ltd	Contoso	Economy	25.49	50	887	22
7	Contoso Bluetooth Not...	Transceiver-free wireles...	Contoso, Ltd	Contoso	Economy	25.49	50	888	22
8	Contoso Bluetooth Not...	Transceiver-free wireles...	Contoso, Ltd	Contoso	Economy	25.49	50	889	22
9	Contoso Bluetooth Not...	Transceiver-free wireles...	Contoso, Ltd	Contoso	Economy	25.49	50	890	22
10	Contoso ADSL Modem ...	ADSL modem splitter/fil...	Contoso, Ltd	Contoso	Economy	9.13	17.9	833	22
11	Contoso ADSL Modem ...	ADSL modem splitter/fil...	Contoso, Ltd	Contoso	Economy	11.68	22.9	834	22

Query 1 +

9 columns, 1,690 rows

8.21 – Enter the following query with DISTINCT with a column reference

EVALUATE

DISTINCT('Product'[Manufacturer])

Notice this also looks like ALL and VALUES at this point.

Results | Result 1 of 1 ▾ Copy ▾

	Product[Manufacturer]
1	Contoso, Ltd
2	Wide World Importers
3	Northwind Traders
4	Adventure Works
5	Southridge Video
6	Litware, Inc.
7	Proseware, Inc.
8	A. Datum Corporation
9	The Phone Company
10	Fabrikam, Inc.

Query 1 +

ess (27.7 ms) Query 1 of 1 Result 1 of 1 1 column, 10 rows

Note - ALL is the only column reference that allows for more than one column reference when creating a table. Both VALUES and DISTINCT can only reference one column in a formula.

Section 8.6 - Understanding Blank Rows in Power BI Visuals

What They Are and Why They Appear

Blank rows in Power BI visuals are placeholder rows that appear in tables or charts, often labeled as "(Blank)". These rows can show up in your visuals when certain DAX functions or data model configurations lead to missing, unmatched, or filtered-out data entries.

Why Do Blank Rows Exist?

Blank rows typically appear for the following reasons:

- **Unmatched Relationships:** When you create a relationship between tables (such as between a Sales table and a Product table), but there are entries in the fact table (e.g., Sales) that do not have a corresponding match in the dimension table (e.g., Product), Power BI inserts a blank row. This is sometimes called the "unknown" member.
- **ALL, VALUES, and DISTINCT Functions:** When using DAX functions like ALL, VALUES, or DISTINCT on columns, especially after filtering or creating new tables, blank rows can be introduced. For example, filtering out a brand such as "Contoso" from a Product table and then using VALUES or ALL on the resulting table may result in a blank row.
- **Filter Context and Missing Data:** If a filter removes all known data for a particular category, or if data is missing or deleted, the blank row acts as a catch-all to ensure the visual can still display totals or subtotals for records not otherwise matched.

We will create a new table with missing data to create a relationship with missing keys which will show how blank rows are handled.

8.22 – Switch to table view and create a new table for all products except Contoso using the following formula.

Product Table No Contoso = `FILTER('Product','Product'[BrandName]<>"Contoso")`

8.23 - Create a relationship between Sales and New Product Table using the [ProductKey] columns in each table.

New relationship



Select tables and columns that are related.

From table

Product Table No Contoso

BrandName	ClassName	Manufacturer	ProductDescri...	ProductKey	ProductName	ProductSu
Wide World I...	Regular	Wide World I...	Record and li...	49	WWI 2GB Pul...	4
Wide World I...	Regular	Wide World I...	Record and li...	50	WWI 2GB Pul...	4
Wide World I...	Regular	Wide World I...	Record and li...	51	WWI 2GB Pul...	4

To table

Sales

channelKey	DateKey	DiscountQua...	ProductKey	PromotionKey	ReturnQuantity	SalesKey
1	5/7/2011 12:0...	0	375	1	0	2201
1	5/22/2013 12:...	0	375	1	0	26527
1	9/30/2013 12:...	0	375	1	0	26702

Cardinality

One to many (1:*)

Cross-filter direction

Single

☒ Make this relationship active

☐ Apply security filter in both directions

8.24 – Switch to DAX Query view and enter the following query

EVALUATE

`ALL('Product Table No Contoso'[BrandName])`

Notice this returns 9 values (No Contoso) and includes a blank row. This is because the sales table has values for Contoso that doesn't match a value in the 'Product Table No Contoso' table.

Results | Result 1 of 1  **Copy** 

	Product Table No Conto...
1	Wide World Importers
2	Adventure Works
3	Southridge Video
4	Litware
5	A. Datum
6	The Phone Company
7	Fabrikam
8	Proseware
9	Northwind Traders
10	

		 Query 1	
Access (143.2 ms)		Query 1 of 1	Result 1 of 1 1 column, 10 rows

8.25 – Update the query replacing ALL with VALUES

EVALUATE

`VALUES('Product Table No Contoso'[BrandName])`

Notice this looks exactly like the ALL result above.

8.26 – Update the query replacing VALUES with DISTINCT
EVALUATE

DISTINCT('Product Table No Contoso'[BrandName])

Results

Result 1 of 1

Copy

	Product Table No Conto...
1	Wide World Importers
2	Adventure Works
3	Southridge Video
4	Litware
5	A. Datum
6	The Phone Company
7	Fabrikam
8	Proseware
9	Northwind Traders



Query 1

Query 1 of 1 Result 1 of 1 1 column, 9 rows

DISTINCT only returns 9 rows with no blank row

8.27 – Let's see this behavior in a visual. Create a table visual using 'Sales'[SalesAmount] with 'Product'[Manufacturer] and a second table visual using 'Sales'[Revenue] with 'Product Table No Contoso'[Manufacturer].

Manufacturer	SalesAmount
A. Datum Corporation	\$619,803,753.56
Adventure Works	\$1,089,734,248.74
Contoso, Ltd	\$1,497,920,768.36
Fabrikam, Inc.	\$1,874,455,854.35
Litware, Inc.	\$327,894,810.44
Northwind Traders	\$20,163,003.97
Proseware, Inc.	\$954,540,220.53
Southridge Video	\$471,653,445.03
The Phone Company	\$673,525,407.96
Wide World Importers	\$811,532,851.91
Total	\$8,341,224,364.83

Manufacturer	SalesAmount
	\$1,497,920,768.36
A. Datum Corporation	\$619,803,753.56
Adventure Works	\$1,089,734,248.74
Fabrikam, Inc.	\$1,874,455,854.35
Litware, Inc.	\$327,894,810.44
Northwind Traders	\$20,163,003.97
Proseware, Inc.	\$954,540,220.53
Southridge Video	\$471,653,445.03
The Phone Company	\$673,525,407.96
Wide World Importers	\$811,532,851.91
Total	\$8,341,224,364.83

Notice the blank row SalesAmount matches the SalesAmount for Contoso

Section 8.7 – Modify tables with SUMMARIZE, ADDCOLUMNS or SUMMARIZECOLUMNS

What if we want to modify a table by selecting specific columns and deduplicating the values? Perhaps even creating new columns from values? This is where we can use SUMMARIZE, ADDCOLUMNS or SUMMARIZECOLUMNS

8.28 – In DAX query view enter the following query

EVALUATE

```
SUMMARIZE('Product','Product'[BrandName],'Product'[ClassName])
```

Results | Result 1 of 1 | Copy

	Product[BrandName]	Product[ClassName]
1	Contoso	Economy
2	Wide World Importers	Economy
3	Northwind Traders	Economy
4	Adventure Works	Economy
5	Southridge Video	Economy
6	Litware	Economy
7	Proseware	Economy
8	A. Datum	Economy
9	The Phone Company	Economy
10	Fabrikam	Economy
11	Contoso	Regular
12	Wide World Importers	Regular

Query 1 +

44.2 ms | Query 1 of 1 | Result 1 of 1 | 2 columns, 29 rows

This creates a new table with unique combinations from both columns

8.29 - ADDCOLUMNS is a way to add a new column into the newly created table. We introduce CALCULATE below and will explain how it works shortly. Enter the following query

EVALUATE

```
ADDCOLUMNS(SUMMARIZE('Product','Product'[BrandName],'Product'[ClassName]),"Sales Amount",CALCULATE(SUM('Sales'[SalesAmount])))
```

Results | Result 1 of 1 Copy

	Product[BrandName]	Product[ClassName]	[Unit Price]
1	Contoso	Economy	277868875.39
2	Wide World Importers	Economy	191553277.11
3	Northwind Traders	Economy	11551306.53
4	Adventure Works	Economy	148905083.74
5	Southridge Video	Economy	129786863.57
6	Litware	Economy	51517141.64
7	Proseware	Economy	101011593.31
8	A. Datum	Economy	97045726.08
9	The Phone Company	Economy	48025341.1
10	Fabrikam	Economy	150969141.35
11	Contoso	Regular	840891052.92
12	Wide World Importers	Regular	118020738.04

Query 1

Success (37.3 ms) | Query 1 of 1 | Result 1 of 1 | 3 columns, 29 rows

Notice this created a new column that aggregates the SalesAmount for each combination.

8.30 – Enter the following query using SUMMARIZECOLUMNS. This acts as a combination of SUMMARIZE and ADDCOLUMNS.

EVALUATE

```
SUMMARIZECOLUMNS('Product'[BrandName], 'Product'[ClassName], "Sales Amount", CALCULATE(SUM(Sales[SalesAmount])))
```

Results | Result 1 of 1 Copy

	Product[BrandName]	Product[ClassName]	[Sales Amount]
1	Contoso	Economy	277868875.39
2	Wide World Importers	Economy	191553277.11
3	Northwind Traders	Economy	11551306.53
4	Adventure Works	Economy	148905083.74
5	Southridge Video	Economy	129786863.57
6	Litware	Economy	51517141.64
7	Proseware	Economy	101011593.31
8	A. Datum	Economy	97045726.08
9	The Phone Company	Economy	48025341.1
10	Fabrikam	Economy	150969141.35
11	Contoso	Regular	840891052.92
12	Wide World Importers	Regular	118030738.01

Query 1

ess (28.1 ms) Query 1 of 1 Result 1 of 1 3 columns, 29 rows

Section 8.8 – Summary Table

FUNCTION	USE WITH COLUMN	USE WITH TABLE	USE WITH MULTIPLE COLUMNS	CREATE NEW COLUMNS	NOTES	FILTER CONTEXT
VALUES	YES	YES	NO	NO	KEEPS BLANK ROW	FOLLOWS
DISTINCT	YES	YES	NO	NO	IGNORES BLANK ROW	FOLLOWS
ALL	YES	YES	YES	NO	KEEPS BLANK ROW	IGNORES
SUMMARIZE	YES	YES	YES	NO		
ADDCOLUMNS				YES		